MECHANICS APPLIED TO AEROSPACE ENGINEERING — PART I

Topic 4: Angular Momentum and Central Forces (Kepler's Problem)

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Sources: slides/04_-_Angular_momentum.pdf & teoria/Notes.pdf (Section 4.8, Chapter 8). Equation numbers are those printed in Notes.pdf.

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1. Angular Momentum and Torque

WHY INTRODUCE A NEW QUANTITY?

Newton's second law gives the evolution of the linear momentum $\mathbf{p}_0 = m\mathbf{v}_0^P$. In many problems the forces have a simple *moment* about a point (for example, a force that always points to a fixed centre has no moment about that centre), and the quantity whose evolution is governed by the moment of the force is the moment of the momentum. This is the angular momentum (Slide 3, Notes Sec. 4.8).

1.1 Definition and Properties

For a particle P of mass m moving with velocity $\mathbf{v}_0^P$ in an inertial frame S_0, the angular momentum about an arbitrary point A (fixed or moving) is

$$\mathbf{H}_{A,0}^P = \mathbf{r}_A^P \times m\mathbf{v}_0^P = \left(\mathbf{r}_0^P - \mathbf{r}_0^A\right) \times m\mathbf{v}_0^P \qquad \text{(Notes Eq. 4.48)}$$

$$\mathbf{r}_A^P = \mathbf{AP} \text{ is the position of } P \text{ relative to } A.$$

- **Direction:** $\mathbf{H}_{A,0}^P$ is perpendicular to both $\mathbf{AP}$ and $\mathbf{v}_0^P$, because it is their cross product.

- **Units:** $[\mathbf{H}] = \mathrm{kg}\cdot\mathrm{m}^2/\mathrm{s}$.
- **Dependence on the point.** For another point B, write $\mathbf{BP} = \mathbf{BA} + \mathbf{AP}$:

$$\mathbf{H}^P_{B,0} = (\mathbf{BA} + \mathbf{AP}) \times m\mathbf{v}^P_0 = \mathbf{H}^P_{A,0} + \mathbf{BA} \times m\mathbf{v}^P_0$$

The value also depends on the frame, because $\mathbf{v}^P_0$ is measured in S_0.

1.2 Torque (Moment of a Force)

If $\mathbf{F}$ is the resultant force on P, its moment about A is the **torque** (Notes Eq. 4.51)

$$\mathbf{M}_A = \mathbf{AP} \times \mathbf{F} = \left(\mathbf{r}^P_0 - \mathbf{r}^A_0\right) \times \mathbf{F}, \qquad [\mathbf{M}_A] = \mathrm{N}\cdot\mathrm{m}$$

Line of action. Split $\mathbf{AP} = \mathbf{AP}_{\parallel} + \mathbf{AP}_{\perp}$ into the parts parallel and perpendicular to $\mathbf{F}$. Then $\mathbf{AP}_{\parallel} \times \mathbf{F} = \mathbf{0}$ and $\mathbf{M}_A = \mathbf{AP}_{\perp} \times \mathbf{F}$, with modulus $d\,\|\mathbf{F}\|$, where $d = \|\mathbf{AP}_{\perp}\|$ is the distance from A to the line of action. Sliding the force along its line does not change the torque.

1.3 Action and Reaction

Two particles P and Q interact with forces $\mathbf{F}^P$ on P and $\mathbf{F}^Q = -\mathbf{F}^P$ on Q, directed along the line PQ. About any point A:

$$\mathbf{M}^P_A + \mathbf{M}^Q_A = \mathbf{AP} \times \mathbf{F}^P - \mathbf{AQ} \times \mathbf{F}^P = (\mathbf{AP} - \mathbf{AQ}) \times \mathbf{F}^P = \mathbf{QP} \times \mathbf{F}^P = \mathbf{0}$$

because $\mathbf{QP}$ is parallel to $\mathbf{F}^P$. Hence internal torques cancel in pairs and only external torques act on a system as a whole (Slide 3).

2. The Angular Momentum Theorem with a Moving Origin

> WHY THIS EQUATION?
> We want the rate of change of $\mathbf{H}^P_{A,0}$. It is obtained by differentiating the definition and using Newton's second law; nothing is assumed about how A moves, so a term appears that vanishes only in special cases (Slide 4, Notes Eqs. 4.49–4.52).

2.1 Derivation

Step 1: product rule on $\left(\mathbf{r}^P_0 - \mathbf{r}^A_0\right) \times m\mathbf{v}^P_0$ with the derivative taken in S_0:

$$\left.\frac{d\mathbf{H}_{A,0}^{P}}{dt}\right|_{0} = \left(\mathbf{v}_0^P - \mathbf{v}_0^A\right) \times m\mathbf{v}_0^P + \left(\mathbf{r}_0^P - \mathbf{r}_0^A\right) \times m\mathbf{a}_0^P \qquad \text{(Eq. 4.49)}$$

Step 2: cross product of a vector with itself. $\mathbf{v}_0^P \times \mathbf{v}_0^P = \mathbf{0}$, so the first term reduces to $-\mathbf{v}_0^A \times m\mathbf{v}_0^P$ (Eq. 4.50).

Step 3: Newton's second law $m\mathbf{a}_0^P = \mathbf{F}$ turns the second term into $\mathbf{AP} \times \mathbf{F} = \mathbf{M}_A$ (Eq. 4.51). Therefore

$$\boxed{\left.\frac{d\mathbf{H}_{A,0}^{P}}{dt}\right|_{0} = \mathbf{M}_A - \mathbf{v}_0^A \times m\mathbf{v}_0^P} \qquad \text{(Eq. 4.52)}$$

2.2 Special Cases

- A **fixed in** S_0 ($\mathbf{v}_0^A = \mathbf{0}$), or $\mathbf{v}_0^A \parallel \mathbf{v}_0^P$ (so that $\mathbf{v}_0^A \times \mathbf{v}_0^P = \mathbf{0}$):

$$\left.\frac{d\mathbf{H}_{A,0}^{P}}{dt}\right|_{0} = \mathbf{M}_A \qquad \text{(Eq. 4.53)}$$

  This is the rotational analogue of $\dot{\mathbf{p}}_0 = \mathbf{F}$.

- A **moves with constant velocity.** Use the auxiliary frame $S_A : \{A; \mathcal{B}_0\}$ with parallel axes. There is no rotation (the change-of-basis matrix is the identity), so $\mathbf{v}_0^P = \mathbf{v}_0^A + \mathbf{v}_A^P$ and $\mathbf{a}_0^P = \mathbf{a}_0^A + \mathbf{a}_A^P$ (Notes Sec. 4.8, text before Eq. 4.54). If $\mathbf{a}_0^A = \mathbf{0}$, then $\mathbf{a}_A^P = \mathbf{a}_0^P$ and, with $\mathbf{H}_{A,A} = \mathbf{AP} \times m\mathbf{v}_A^P$,

$$\left.\frac{d\mathbf{H}_{A,A}}{dt}\right|_{A} = \mathbf{v}_A^P \times m\mathbf{v}_A^P + \mathbf{AP} \times m\mathbf{a}_A^P = \mathbf{AP} \times \mathbf{F} = \mathbf{M}_A \qquad \text{(Eq. 4.55)}$$

  because the derivative of $\mathbf{AP}$ in S_A is $\mathbf{v}_A^P$ and $\mathbf{v}_A^P \times \mathbf{v}_A^P = \mathbf{0}$. This is the reason why the barycentre of a two-body system (which moves uniformly, Section 5) can be used as the origin.

> **i NO NEW PHYSICS**
> The angular momentum equation is a consequence of Newton's second law, so it adds no information that the force equation did not already contain (Notes, after Eq. 4.53). Its use is that it gives first integrals directly and projects naturally on any basis.

3. Conservation of Angular Momentum

Single particle. If A is fixed and $\mathbf{M}_A = \mathbf{0}$ at all times, Eq. 4.53 gives $d\mathbf{H}^P_{A,0}/dt = \mathbf{0}$, so $\mathbf{H}^P_{A,0}$ is a constant vector: three scalar first integrals

$$H_x = m(y\dot{z} - z\dot{y}), \qquad H_y = m(z\dot{x} - x\dot{z}), \qquad H_z = m(x\dot{y} - y\dot{x})$$

If only the component of $\mathbf{M}_A$ along a fixed direction $\mathbf{n}$ vanishes, then $\mathbf{H}^P_{A,0} \cdot \mathbf{n}$ is conserved (Notes, text after Eq. 4.53); the remaining components are not.

System of particles. Adding $\mathbf{H}_{A,0} = \sum_i \mathbf{AP}_i \times m_i \mathbf{v}_0^{P_i}$ and differentiating about a fixed A gives the sum of the torques of all forces; by Section 1.3 the internal torques cancel in pairs, so $d\mathbf{H}_{A,0}/dt = \mathbf{M}^{\text{ext}}_A$. An isolated system conserves its angular momentum (Slide 5).

4. Central Force Motion

> WHY STUDY CENTRAL FORCES SEPARATELY?
> Gravity from a body treated as a point is a central force, and the torque of a central force about its centre is zero by construction. The conservation law of Section 3 then applies and gives, without solving any equation of motion, strong information about the trajectory (Slide 7, Notes Sec. 4.8.1).

A force is **central** about a fixed point O if its line of action always passes through O:

$$\mathbf{F} = F\,\mathbf{e}_r, \qquad \mathbf{e}_r = \frac{\mathbf{r}_0^P}{r} \qquad \text{(Eq. 4.56)}$$

Torque and angular momentum. With $\mathbf{r}_0^P = r\mathbf{e}_r$ taken from the fixed point O:

$$\mathbf{M}_O = \mathbf{r}_0^P \times \mathbf{F} = rF\,(\mathbf{e}_r \times \mathbf{e}_r) = \mathbf{0} \;\; \text{(Eq. 4.57)} \;\Longrightarrow\; \mathbf{H}_{O,0} = \mathbf{r}_0^P \times m\mathbf{v}_0^P = \text{const} \;\; \text{(Eq. 4.58)}$$

4.1 Properties

1. **Three scalar conservation laws**, from $\mathbf{H}_{O,0} = \text{const}$.
2. **The motion is planar.** By definition $\mathbf{H}_{O,0}$ is perpendicular to $\mathbf{r}_0^P$ and $\mathbf{v}_0^P$: $\mathbf{r} \cdot \mathbf{H} = \mathbf{r} \cdot (\mathbf{r} \times m\mathbf{v}) = 0$ and $\mathbf{v} \cdot \mathbf{H} = 0$. A fixed vector $\mathbf{H} \neq \mathbf{0}$ therefore keeps P in the plane through O perpendicular to $\mathbf{H}$. (If $\mathbf{H} = \mathbf{0}$, $\mathbf{r}$ and $\mathbf{v}$ are parallel and the motion is rectilinear through O.)
3. **Polar coordinates in the plane.** Choosing $\mathbf{k}_0$ along $\mathbf{H}$, the motion lies in $z = 0$ and the natural coordinates are (r, θ) (Section 6).

4. **Speed-radius relation.** In polar coordinates $H = mr^2\dot\theta = \text{const}$ (derived in Section 6), so $\dot\theta = h/r^2$ increases when r decreases.
5. **The centre is not reached** if $h \neq 0$, for a sufficiently weak attraction. The transverse kinetic energy is $\frac{1}{2}m(r\dot\theta)^2 = \frac{1}{2}mh^2/r^2 \to \infty$ as $r \to 0$, which for the Newtonian potential $-\mu/r$ cannot be supplied by the potential energy. This is not true for every central force: in Problem 42 the attractions $F \propto 1/r^3$ (with $h^2 < \mu$) and $F \propto 1/r^4$ carry particles with $h \neq 0$ to $r = 0$ (Section 11).

5. Kepler's Problem: Reduction of the Two-Body Problem

5.1 Kepler's Empirical Laws (Slide 8, Notes Sec. 8.1)

1. The orbits of the planets are ellipses with the Sun at one focus.
2. The line joining a planet to the Sun sweeps out equal areas in equal times.
3. The square of the period is proportional to the cube of the major axis: $\tau^2 \propto a^3$.

Kepler gave no explanation. The goal of the next sections is to derive the three laws from Newton's second law and the law of universal gravitation $F_g = Gm^P m^S/d^2$.

> WHY REDUCE THE PROBLEM?
> Two bodies attract each other, so each equation of motion contains both position vectors. Notes Chapter 8 skips this step by assuming $m^P \ll m^S$ and placing the origin at the central body (Eq. 8.1); Slides 9–11 give the general reduction, which is shown here. Adding the two equations eliminates the interaction and gives the motion of the barycentre; subtracting them isolates the relative motion, which is the only part that determines the orbit.

5.2 Equations of Motion in S_0

For the planet or satellite P and the central body S, with $\mathbf{r} \equiv \mathbf{r}_0^P - \mathbf{r}_0^S$ and $r = \|\mathbf{r}\|$:

$$m^P\ddot{\mathbf{r}}_0^P = -\frac{Gm^Pm^S}{r^3}\,\mathbf{r}, \qquad m^S\ddot{\mathbf{r}}_0^S = +\frac{Gm^Pm^S}{r^3}\,\mathbf{r}$$

5.3 Sum: the Barycentre Moves Uniformly

Define $M_{\text{tot}} = m^P + m^S$ and the barycentre $\mathbf{r}_0^G = \left(m^P\mathbf{r}_0^P + m^S\mathbf{r}_0^S\right)/M_{\text{tot}}$. Adding the equations, the forces cancel:

$$m^P\ddot{\mathbf{r}}_0^P + m^S\ddot{\mathbf{r}}_0^S = M_{\text{tot}}\,\ddot{\mathbf{r}}_0^G = \mathbf{0} \implies \mathbf{r}_0^G(t) = \mathbf{r}_0^G(0) + \mathbf{v}_0^G\,t$$

Two successive integrations give a uniform straight-line motion. A frame S_G with origin at G and axes parallel to S_0 differs from S_0 by a translation at constant velocity (change-of-basis matrix equal to the identity, $\det = 1$), so S_G is also inertial and $d/dt|_G = d/dt|_0$.

5.4 Difference: the Relative Motion

Divide the first equation by m^P, the second by m^S, and subtract:

$$\ddot{\mathbf{r}} = \ddot{\mathbf{r}}_0^P - \ddot{\mathbf{r}}_0^S = -\frac{Gm^S}{r^3}\mathbf{r} - \frac{Gm^P}{r^3}\mathbf{r} = -\frac{G\left(m^S + m^P\right)}{r^3}\mathbf{r}$$

With the **standard gravitational parameter** $\mu \equiv G(m^S + m^P)$, the relative motion obeys the one-body Kepler equation (Slide 11):

$$\boxed{\frac{d^2\mathbf{r}}{dt^2} = -\frac{\mu}{r^3}\,\mathbf{r} = -\frac{\mu}{r^2}\,\mathbf{e}_r}$$

5.5 Reduced Mass

Multiplying the last equation by $m_{\text{red}} \equiv m^P m^S/(m^P + m^S)$ and using $m_{\text{red}}\, G(m^S + m^P) = Gm^P m^S$ gives $m_{\text{red}}\,\ddot{\mathbf{r}} = -Gm^P m^S\,\mathbf{e}_r/r^2$: the relative vector moves like a single particle of mass m_{red} attracted to a fixed centre by the full gravitational force. After solving for $\mathbf{r}(t)$, each body is recovered from

$$\mathbf{r}_0^P = \mathbf{r}_0^G + \frac{m^S}{M_{\text{tot}}}\,\mathbf{r}, \qquad \mathbf{r}_0^S = \mathbf{r}_0^G - \frac{m^P}{M_{\text{tot}}}\,\mathbf{r}$$

which follows from the definition of $\mathbf{r}_0^G$ and $\mathbf{r} = \mathbf{r}_0^P - \mathbf{r}_0^S$. Both bodies describe similar conics about G.

Aerospace limit. If $m^S \gg m^P$ (satellite around the Earth), $m_{\text{red}} \to m^P$, $G \approx S$ and $\mu \approx Gm^S$ (Notes Eq. 8.1). For the Earth, $\mu_\oplus = GM_\oplus \approx 3.986004418 \times 10^{14}\ \text{m}^3/\text{s}^2$. The mass of P does not appear in the equation, so mass-specific quantities (per unit mass) are used from now on.

The reduced equation is a **central force problem**: Section 4 applies, so $\mathbf{H}$ is conserved and the motion is planar. The force is also conservative with specific potential energy $V_0 = -\mu/r$ (Section 10).

6. Polar Kinematics, the Constant h, and Kepler's Second Law

> WHY A ROTATING BASIS?
> The force always points along $\mathbf{r}$, so its components are simplest along the direction of $\mathbf{r}$ and the perpendicular direction. Those directions rotate with P, so we attach to P the polar basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$ and compute its time derivatives (Notes Eqs. 8.2–8.3, Slide 12).

6.1 Change of Basis and Poisson Formula

With $\mathbf{k}_0$ along $\mathbf{H}$, $\mathcal{B}_1$ is obtained from $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ by a rotation θ about $\mathbf{k}_0$:

$$\begin{pmatrix} \mathbf{e}_r \\ \mathbf{e}_\theta \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}, \qquad [{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Columns of $[{}_0R_1]$: components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0$ in $\mathcal{B}_0$ (Notes Eq. 8.2).

Checks. $\det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1$. Each column has unit modulus and $\mathbf{e}_r \cdot \mathbf{e}_\theta = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$, so $[{}_0R_1]^T[{}_0R_1] = I$.

The angular velocity of $\mathcal{B}_1$ with respect to $\mathcal{B}_0$ is $\boldsymbol{\omega}_{10} = \dot\theta\,\mathbf{k}_0$, and Poisson's formula $\dot{\mathbf{e}} = \boldsymbol{\omega}_{10} \times \mathbf{e}$ gives

$$\dot{\mathbf{e}}_r = \dot\theta\,\mathbf{k}_0 \times \mathbf{e}_r = \dot\theta\,\mathbf{e}_\theta, \qquad \dot{\mathbf{e}}_\theta = \dot\theta\,\mathbf{k}_0 \times \mathbf{e}_\theta = -\dot\theta\,\mathbf{e}_r$$

6.2 Velocity and Acceleration

Differentiate $\mathbf{r} = r\mathbf{e}_r$ with the product rule:

$$\mathbf{v} = \dot r\,\mathbf{e}_r + r\dot{\mathbf{e}}_r = \dot r\,\mathbf{e}_r + r\dot\theta\,\mathbf{e}_\theta$$

$$\mathbf{a} = \ddot r\,\mathbf{e}_r + \dot r\dot\theta\,\mathbf{e}_\theta + \left(\dot r\dot\theta + r\ddot\theta\right)\mathbf{e}_\theta + r\dot\theta\left(-\dot\theta\,\mathbf{e}_r\right) = \left(\ddot r - r\dot\theta^2\right)\mathbf{e}_r + \left(r\ddot\theta + 2\dot r\dot\theta\right)\mathbf{e}_\theta \qquad \text{(Eq. 8.3)}$$

6.3 Equations of Motion and the Constant h

Newton's law $\mathbf{a} = -(\mu/r^2)\,\mathbf{e}_r$ (Eq. 8.4), projected on $\mathbf{e}_r$ and $\mathbf{e}_\theta$:

$$\ddot r - r\dot\theta^2 = -\frac{\mu}{r^2} \quad \text{(Eq. 8.5)}, \qquad r\ddot\theta + 2\dot r\dot\theta = 0 \quad \text{(Eq. 8.6)}$$

Why multiply by r. The product $r\,(r\ddot\theta + 2\dot r\dot\theta) = r^2\ddot\theta + 2r\dot r\dot\theta$ is the total derivative $\dfrac{d}{dt}\left(r^2\dot\theta\right)$ (product rule: $\dfrac{d}{dt}(r^2) = 2r\dot r$). Then Eq. 8.6 reads $\dfrac{d}{dt}(r^2\dot\theta) = 0$ and

$$\boxed{h = r^2\dot{\theta} = \text{const}} \qquad \text{(Eq. 8.7)}$$

h is the modulus of the mass-specific angular momentum, $h = \|\mathbf{r}\times\mathbf{v}\|$, in m^2/s; its value comes from the initial conditions.

6.4 Kepler's Second Law (Areal Velocity)

In dt the radius vector turns by $d\theta$ and sweeps a thin triangle of base $r\,d\theta$ and height r, of area $dA = \tfrac{1}{2}r^2\,d\theta$. Dividing by dt:

$$\frac{dA}{dt} = \frac{1}{2}r^2\dot{\theta} = \frac{h}{2} = \text{const} \qquad \text{(Eq. 8.8)}$$

Equal areas are swept in equal times: this is Kepler's second law. It holds for any central force, not only for gravity.

7. The Binet Equation

> WHY CHANGE VARIABLES?
> Eq. 8.5 is nonlinear in $r(t)$. We look for the shape $r(\theta)$ of the orbit rather than the time law. Because $\dot{\theta} = h/r^2$ has a fixed sign, θ is a monotonic function of t and can replace it as independent variable; with $u = 1/r$ the equation becomes linear (Slide 13, Notes Eqs. 8.9–8.12).

Define $u(\theta) \equiv 1/r(\theta)$ and write $' = d/d\theta$. From Eq. 8.7, $\dot{\theta} = h/r^2 = hu^2$, so the time derivative of any function of θ is $\dot{f} = f'\,\dot{\theta} = hu^2 f'$.

7.1 Derivatives of r

Step 1: chain rule for $r = u^{-1}$. Since $\dfrac{d}{du}u^{-1} = -u^{-2}$,

$$\frac{dr}{d\theta} = \frac{d\,(1/u)}{d\theta} = \frac{d\left(u^{-1}\right)}{du}\,\frac{du}{d\theta} = -\frac{u'}{u^2}$$

Step 2: radial velocity.

$$\dot{r} = \frac{dr}{d\theta}\,\dot{\theta} = \left(-\frac{u'}{u^2}\right)\left(hu^2\right) = -h\,u' \qquad \text{(Eq. 8.10)}$$

Step 3: radial acceleration. Differentiate $\dot r = -h\,u'(\theta)$ in time with the chain rule (h is constant):

$$\ddot r = \frac{d}{dt}\left(-h\,u'\right) = -h\,u''\,\dot\theta = -h\,u''\left(hu^2\right) = -h^2u^2\,u'' \qquad \text{(Eq. 8.11)}$$

Step 4: centrifugal term.

$$r\dot\theta^2 = \frac{1}{u}\left(hu^2\right)^2 = h^2u^3$$

7.2 Binet Equation for the Inverse-Square Force

Insert Steps 3 and 4 in Eq. 8.5, with $-\mu/r^2 = -\mu u^2$:

$$-h^2u^2\,u'' - h^2u^3 = -\mu u^2 \quad \overset{\div\,(-h^2u^2)}{\Longrightarrow} \quad \boxed{\frac{d^2u}{d\theta^2} + u = \frac{\mu}{h^2}} \qquad \text{(Eq. 8.12)}$$

The division is allowed because $u \neq 0$ at finite distance. This is a forced harmonic oscillator in the variable θ. For a general central force $F(r)\mathbf{e}_r$ the same steps give $u'' + u = -F(1/u)/(mh^2u^2)$, which is the form used for the power-law force of Problem 42.

8. Conic Trajectories and Orbit Geometry

8.1 Solution of the Binet Equation

1. **Homogeneous equation** $u_h'' + u_h = 0$: characteristic roots $\lambda = \pm i$, so $u_h = C_1\cos\theta + C_2\sin\theta$.
2. **Particular solution:** a constant $u_p = c$ gives $c = \mu/h^2$.
3. **General solution** (Eq. 8.13):

$$u(\theta) = \frac{\mu}{h^2} + C_1\cos\theta + C_2\sin\theta$$

Constants from the initial data. At the initial angle (taken as $\theta = 0$ for this step) the particle has $u = u_i = 1/r_i$ and, from $\dot r = -hu'$, $u' = u_i' = -\dot r_i/h$. Then

$$u(0) = \frac{\mu}{h^2} + C_1 = u_i \;\Longrightarrow\; C_1 = u_i - \frac{\mu}{h^2}, \qquad u'(0) = C_2 = u_i'$$

Write $C_1\cos\theta + C_2\sin\theta = A\cos(\theta - \theta_0)$ with $A = \sqrt{C_1^2 + C_2^2}$, $A\cos\theta_0 = C_1$, $A\sin\theta_0 = C_2$, and define the **eccentricity**

$$e \equiv \frac{Ah^2}{\mu} = \frac{h^2}{\mu}\sqrt{\left(u_i - \frac{\mu}{h^2}\right)^2 + u_i'^2} \geq 0, \qquad u(\theta) = \frac{\mu}{h^2}\left[1 + e\cos(\theta - \theta_0)\right]$$

Choice of the polar axis. The closest point to the focus is where u is maximum: at $\theta = \theta_0$ one has $u' = -(\mu e/h^2)\sin(\theta - \theta_0) = 0$ and $u'' = -(\mu e/h^2) < 0$. Redefine the polar axis so that $\theta = 0$ points to this periapsis, which means $\theta_0 = 0$ (a rotation of the plane basis by θ_0, with the same matrix $[{}_0R_1]$ as in Section 6.1, $\det = 1$, $R^T R = I$). Then

$$\frac{1}{r} = \frac{\mu}{h^2}\left(1 + e\cos\theta\right) \quad\Longrightarrow\quad \boxed{r(\theta) = \frac{h^2/\mu}{1 + e\cos\theta} = \frac{p}{1 + e\cos\theta}}, \qquad p \equiv \frac{h^2}{\mu} \qquad \text{(Eq. 8.14)}$$

This is the polar equation of a **conic section with one focus at O**, with semi-latus rectum p and eccentricity e fixed by the initial position and velocity. For $e < 1$ the orbit is an ellipse with the central body at a focus: Kepler's first law.

Eccentricity	Curve	Semi-major axis a	Bound?
$e = 0$	Circle	$a = p > 0$	Bound
$0 < e < 1$	Ellipse	$a > 0$	Bound
$e = 1$	Parabola	$a \to \infty$	Escape threshold
$e > 1$	Hyperbola	$a < 0$	Unbound

8.2 Geometry of the Ellipse (Eqs. 8.14–8.16)

Evaluate $r(\theta)$ at the two apsides, $\theta = 0$ and $\theta = \pi$:

$$r_p = \frac{p}{1 + e}, \qquad r_a = \frac{p}{1 - e}$$

The major axis is $2a = r_p + r_a$ (Eq. 8.15), hence

$$2a = \frac{p(1 - e) + p(1 + e)}{(1 + e)(1 - e)} = \frac{2p}{1 - e^2} \quad\Longrightarrow\quad p = a\left(1 - e^2\right)$$

so that $r_p = a(1-e)$, $r_a = a(1+e)$. Combining $p = a(1-e^2)$ with $p = h^2/\mu$ (Eq. 8.16):

$$\boxed{h^2 = \mu p = \mu a\left(1-e^2\right), \qquad h = \quad \mu a\left(1-e^2\right)}$$

The distance from the centre of the ellipse to a focus is $c = a - r_p = ae$. At the end of the minor axis the two focal distances are equal and their sum is $2a$, so each is a and $b^2 + c^2 = a^2$, which gives $b = a \quad 1-e^2$. For circles and ellipses $a > 0$, for parabolas $a \to \infty$ and for hyperbolas $a < 0$.

9. Orbital Period and Kepler's Third Law

> WHY THE AREAL VELOCITY?
> The trajectory equation gives $r(\theta)$ but not $\theta(t)$. Over one full orbit, however, the swept area is known (the area of the ellipse) and the swept area per unit time is constant, which fixes the period without solving for $\theta(t)$ (Slide 15, Notes Eqs. 8.16–8.18).

The area of an ellipse is $A_e = \pi ab$. Integrate $dA/dt = h/2$ over one period τ (Barrow's rule on each side):

$$\int_0^{\tau} \frac{h}{2}\,dt = \int_0^{A_e} dA \quad\Longrightarrow\quad \frac{h}{2}\,\tau = \pi ab \quad\Longrightarrow\quad \tau = \frac{2\pi ab}{h}$$

Substitute $b = a \quad 1-e^2$ and $h = \quad \mu a(1-e^2)$ (Section 8.2):

$$\tau = \frac{2\pi a \cdot a \quad 1-e^2}{\mu a \quad 1-e^2} = \frac{2\pi a^{3/2}}{\mu} = 2\pi \quad \frac{a^3}{\mu} \qquad \text{(Eq. 8.18)}, \qquad \tau^2 = \frac{4\pi^2}{\mu}\,a^3 \qquad \text{(Eq. 8.17)}$$

The factor $\quad 1-e^2$ cancels, so the period depends only on a and μ, not on the eccentricity: Kepler's third law. (With $m^S \gg m^P$ the constant $4\pi^2/\mu$ is practically the same for all planets of the same star.)

Kepler's law	Statement	Where it was derived
First	Elliptical orbit with the central body at a focus	Section 8.1, from the Binet equation (also parabolas and hyperbolas for $e \geq 1$)
Second	$dA/dt = h/2 = \text{const}$	Section 6.4, any central force
Third	$\tau^2 = 4\pi^2 a^3/\mu$	Section 9, inverse-square force only

10. Energy and the Vis-Viva Equation

10.1 Specific Energy

Why. The gravitational force is conservative, so mechanical energy is conserved and gives a relation between speed and position that needs no time integration (Slides 11, 16, Notes Sec. 8.5). The potential energy per unit mass follows from the work of the force with Barrow's rule (reference at infinity):

$$V_0(r) = -\int_\infty^r \left(-\frac{\mu}{s^2}\right) ds = \mu\left[-\frac{1}{s}\right]_\infty^r = -\frac{\mu}{r}, \qquad \xi = \frac{1}{2}v^2 - \frac{\mu}{r} = \text{const} \qquad \text{(Eq. 8.19)}$$

10.2 Value of the Constant

At the apsides $\dot r = 0$ (because $u' \propto \sin\theta$ vanishes), so the velocity is purely transverse and $h = rv$ gives v directly.

Periapsis ($r_p = a(1-e)$):

$$v_p = \frac{h}{r_p} = \frac{\mu a(1-e^2)}{a(1-e)} = \frac{\mu}{a}\frac{(1+e)(1-e)}{(1-e)^2} = \frac{\mu}{a}\frac{1+e}{1-e}$$

$$\xi = \frac{v_p^2}{2} - \frac{\mu}{r_p} = \frac{\mu}{2a}\frac{1+e}{1-e} - \frac{\mu}{a(1-e)} = \frac{\mu}{2a(1-e)}\left[(1+e)-2\right] = -\frac{\mu}{2a}$$

Apoapsis ($r_a = a(1+e)$):

$$v_a = \frac{h}{r_a} = \frac{\mu}{a}\frac{1-e}{1+e}, \qquad \xi = \frac{\mu}{2a}\frac{1-e}{1+e} - \frac{\mu}{a(1+e)} = \frac{\mu}{2a(1+e)}\left[(1-e)-2\right] = -\frac{\mu}{2a}$$

Check at any θ. With $u = \frac{\mu}{h^2}(1+e\cos\theta)$, $u' = -\frac{\mu}{h^2}e\sin\theta$, $\dot r = -hu'$ and $r\dot\theta = hu$, the energy is $\xi = \frac{1}{2}h^2\left[(u')^2 + u^2\right] - \mu u$:

$$\xi = \frac{\mu^2}{2h^2}\left[e^2\sin^2\theta + (1+e\cos\theta)^2\right] - \frac{\mu^2}{h^2}(1+e\cos\theta) = \frac{\mu^2}{2h^2}\left(e^2-1\right) = -\frac{\mu}{2a}$$

where the last equality uses $h^2 = \mu a(1-e^2)$. The result is independent of θ, as it must be.

10.3 Vis-Viva Equation and Characteristic Speeds

$$\frac{1}{2}v^2 - \frac{\mu}{r} = -\frac{\mu}{2a} \quad \Longrightarrow \quad \boxed{v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right)} \qquad \text{(Eq. 8.19)}$$

Quantity	Expression	Derivation
Energy sign	$\xi < 0$ ellipse, $\xi = 0$ parabola, $\xi > 0$ hyperbola	$\xi = -\mu/(2a)$ with $a > 0$, $a \to \infty$, $a < 0$
Circular speed	$v_c = \sqrt{\mu/r}$ (Eq. 8.20)	$r = a$ in the vis-viva equation
Escape speed	$v_e = \sqrt{2\mu/r} = \sqrt{2}\, v_c$ (Eq. 8.21)	$\xi = 0$, i.e. $a \to \infty$
Periapsis speed	$v_p = \sqrt{\dfrac{\mu}{a}\dfrac{1+e}{1-e}}$	Section 10.2
Apoapsis speed	$v_a = \sqrt{\dfrac{\mu}{a}\dfrac{1-e}{1+e}}$	Section 10.2
Apsidal ratio	$\dfrac{v_p}{v_a} = \dfrac{1+e}{1-e} = \dfrac{r_a}{r_p}$	$r_p v_p = r_a v_a = h$

For a hyperbola the speed at infinity is $v_\infty = \sqrt{-\mu/a}$, from $\xi = v_\infty^2/2$ and $\xi = -\mu/(2a)$ with $a < 0$.

11. Effective Potential in the Variable u and General Power Laws

> WHY A FIRST INTEGRAL IN U?
>
> For a central force other than the inverse-square law (as in Problem 42), the equation $u'' + u = \cdots$ is not a linear oscillator and cannot be solved in closed form for every exponent. A first integral still tells us where the motion can go (escape, collision with the centre, or oscillation) without solving it. This section extends Notes Chapter 8; Notes.pdf does not discuss it.

11.1 First Integral

Let the central force be conservative with specific potential $V_0(r)$. With $\dot r = -h u'$ and $r\dot\theta = hu$ the specific energy is

$$\xi = \frac{1}{2}\left(\dot r^2 + r^2\dot\theta^2\right) + V_0 = \frac{1}{2}h^2\left[(u')^2 + u^2\right] + V_0\!\left(\frac{1}{u}\right)$$

$$\boxed{(u')^2 + W_{\text{eff}}(u) = \frac{2\xi}{h^2} \equiv E^*}, \qquad W_{\text{eff}}(u) = u^2 + \frac{2V_0(1/u)}{h^2}$$

Since $(u')^2 \geq 0$, motion is possible only where $W_{\text{eff}}(u) \leq E^*$, and turning points ($u' = 0$) are the roots of $W_{\text{eff}}(u) = E^*$. Differentiating the first integral with respect to θ and dividing by $2u'$ gives $u'' + \tfrac{1}{2}W'_{\text{eff}}(u) = 0$. For the Newtonian potential $V_0 = -\mu u$:

$$W_{\text{eff}}(u) = u^2 - \frac{2\mu}{h^2}u = \left(u - \frac{\mu}{h^2}\right)^2 - \left(\frac{\mu}{h^2}\right)^2, \qquad u'' = -\frac{1}{2}W'_{\text{eff}} = -u + \frac{\mu}{h^2}$$

which is the Binet equation again. The minimum at $u = \mu/h^2$ is the circular orbit $r = p$; $E^* < 0$ gives two turning points (ellipse), $E^* \geq 0$ reaches $u = 0$ (escape).

11.2 Power-Law Attraction $F = -m\mu/r^\alpha$

Here $V_0 = -\mu/((\alpha-1)r^{\alpha-1})$ for $\alpha \neq 1$ and $V_0 = \mu \ln r$ for $\alpha = 1$ (potential obtained as in Section 10.1). The Binet equation and effective potential are

$$u'' + u = \frac{\mu}{h^2}\,u^{\alpha-2}, \qquad W_{\text{eff}}(u) = u^2 - \frac{2\mu}{(\alpha-1)h^2}u^{\alpha-1} \;\; (\alpha \neq 1), \qquad W_{\text{eff}}(u) = u^2 - \frac{2\mu}{h^2}\ln u \;\; (\alpha = 1)$$

α	$W_{\text{eff}}(u)$	**Behaviour**
1	$u^2 - \dfrac{2\mu}{h^2}\ln u$	$W_{\text{eff}} \to +\infty$ at both $u \to 0$ and $u \to \infty$: every orbit is bounded and oscillates between two turning points.
2	$u^2 - \dfrac{2\mu}{h^2}u$	Kepler: bounded for $E^* < 0$, escape for $E^* \geq 0$, no collision if $h \neq 0$.
3	$\left(1 - \dfrac{\mu}{h^2}\right)u^2$	$h^2 > \mu$: always escapes. $h^2 < \mu$: collapse or escape depending on u'_0 (sign of C_1). No oscillation.
4	$u^2 - \dfrac{2\mu}{3h^2}u^3$	Minimum only at $u = 0$, barrier $W_{\max} = h^4/(3\mu^2)$ at $u = h^2/\mu$. No bounded oscillation: escape or collapse.

The full case-by-case discussion with initial data (u_0, u'_0) is worked in Problem 42.

11.3 Circular Orbits and Their Stability

A circular orbit has $u' = 0$ and $u'' = 0$, so it sits at a stationary point of W_{eff}: $W'_{\text{eff}}(u_0) = 2u_0 - 2\mu u_0^{\alpha-2}/h^2 = 0$, that is $u_0^{\alpha-3} = h^2/\mu$. The second derivative is $W''_{\text{eff}}(u_0) = 2 - 2(\alpha-2)\,\dfrac{\mu}{h^2}\,u_0^{\alpha-3}$, and inserting $u_0^{\alpha-3} = h^2/\mu$ gives $W''_{\text{eff}}(u_0) = 2 - 2(\alpha-2) = 2(3-\alpha)$. A minimum (stable circular orbit) therefore exists for $\alpha < 3$, and a maximum (unstable) for $\alpha > 3$; $\alpha = 3$ is marginal. This is consistent with Kepler's circular orbit ($\alpha = 2$, minimum) and with the unstable circle of Problem 42 for $\alpha = 4$.

12. Worked Numerical Examples

Example 1: Low Earth Circular Orbit

Circular speed, escape speed and period

Eqs. 8.18, 8.20, 8.21

Data. $\mu_\oplus = 3.986004418 \times 10^{14}\ \mathrm{m^3/s^2}$ and orbital radius $r = 6.778 \times 10^{6}\ \mathrm{m}$ (about 400 km altitude).

1. Circular speed: $v_c = \sqrt{\mu/r} = \sqrt{3.986004418 \times 10^{14}/6.778 \times 10^{6}} \approx 7.669 \times 10^{3}\ \mathrm{m/s}$.
2. Escape speed at the same radius: $v_e = \sqrt{2}\, v_c \approx 1.0845 \times 10^{4}\ \mathrm{m/s}$.
3. Period with $a = r$: $\tau = 2\pi \sqrt{r^3/\mu} \approx 5.553 \times 10^{3}\ \mathrm{s} \approx 92.6\ \mathrm{min}$.
4. Dimensions: $[\mu/r] = \mathrm{m^3 s^{-2}/m} = \mathrm{m^2/s^2}$, so v is in m/s; $[r^3/\mu] = \mathrm{s^2}$, so τ is in s.

Example 2: Elliptic Transfer Orbit (Vis-Viva)

Periapsis and apoapsis from the given apsides

Section 10

Data. Periapsis radius $r_p = 6.778 \times 10^{6}\ \mathrm{m}$ and apoapsis radius $r_a = 4.2164 \times 10^{7}\ \mathrm{m}$ around the Earth.

1. Semi-major axis and eccentricity: $a = (r_p + r_a)/2 = 2.4471 \times 10^{7}\ \mathrm{m}$ and $e = (r_a - r_p)/(r_a + r_p) = 0.7230$.
2. Specific energy: $\xi = -\mu/(2a) \approx -8.144 \times 10^{6}\ \mathrm{J/kg}$.
3. Vis-viva at periapsis: $v_p = \sqrt{\mu(2/r_p - 1/a)} \approx 1.0066 \times 10^{4}\ \mathrm{m/s}$; at apoapsis $v_a = \sqrt{\mu(2/r_a - 1/a)} \approx 1.618 \times 10^{3}\ \mathrm{m/s}$.
4. Consistency: $r_p v_p = r_a v_a = h \approx 6.823 \times 10^{10}\ \mathrm{m^2/s}$, equal to $\sqrt{\mu a(1-e^2)}$, and $v_p/v_a = 6.221 = r_a/r_p$.
5. Period: $\tau = 2\pi \sqrt{a^3/\mu} \approx 3.81 \times 10^{4}\ \mathrm{s} \approx 10.58\ \mathrm{h}$.

13. Source Map

Topic	Notes.pdf (printed pages)	Slides
Angular momentum and torque	Sec. 4.8, Eqs. 4.48 and 4.51 (pp. 49–50)	3
Theorem with a moving origin	Eqs. 4.49–4.55 (pp. 50–51)	4
Conservation	Text after Eq. 4.53 (p. 51)	5
Central forces	Sec. 4.8.1, Eqs. 4.56–4.58 and properties 1–5 (pp. 51–52). Note: Notes p. 52 states unconditionally that P can never reach O unless $H = 0$; this disagrees with Problem 42, where $F \propto 1/r^3$ (with $h^2 < \mu$) and $F \propto 1/r^4$ carry particles with $h \neq 0$ to $r = 0$. The statement holds for the Newtonian $1/r^2$ force.	7
Two-body reduction	Not in the Notes (Ch. 8 assumes $m^P \ll m^S$, Eq. 8.1, p. 71)	8–11
Polar kinematics, h, areal velocity	Eqs. 8.2–8.8 (p. 72)	12
Binet equation and conic orbit	Eqs. 8.9–8.16 (pp. 72–73)	13–14
Period and third law	Eqs. 8.16–8.18 (p. 73)	15
Vis-viva and characteristic speeds	Sec. 8.5, Eqs. 8.19–8.21 (p. 74)	16
Effective potential in u, power laws, circular-orbit stability	Not in the Notes; derived from Eq. 8.19 and used in Problem 42	none

Worked problems (Problems 17, 39, 42, 43, 44) follow the four-phase protocol on the Topic 4 problems page.

← PREVIOUS TOPIC
Topic 3: Constraints and Reaction Forces

PRACTICE PROBLEMS
Topic 4 Problems (5)

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Mechanics Applied Index

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